

$$n x(n) \leftrightarrow -z \frac{dX(z)}{dz}$$

$$\text{proof: } X(z) = \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n}$$

$$\frac{dX(z)}{dz} = \sum_{n=-\infty}^{\infty} (-n) \cdot x(n) \cdot z^{-n-1}$$

$$\Rightarrow -z \cdot \frac{dX(z)}{dz} = \sum_{n=-\infty}^{\infty} n x(n)$$